1) a)

$|z| = \sqrt{4^2 + (-2)^2} = \sqrt{16 + 4} = \sqrt{20} = 2\sqrt{5}$

1) b)

$|w| = \sqrt{(-1)^2 + (3)^2} = \sqrt{1 + 9} = \sqrt{10}$

1) c)

$|2w| = \sqrt{(-2)^2 + (6)^2} = \sqrt{4 + 36} = \sqrt{40} = 2\sqrt{10}$

1) d)

$\frac{z}{w} = \frac{4-2\mathrm{i}}{-1+3\mathrm{i}} \times \frac{-1-3\mathrm{i}}{-1-3\mathrm{i}} = \frac{-4+2\mathrm{i}-12\mathrm{i}-6}{1+9} = -1 - \mathrm{i}$

$\left|\frac{z}{w}\right| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{1 + 1} = \sqrt{2}$

2) a)

$\arg(1 + \mathrm{i}) = \arctan\left(\frac{1}{1}\right) = \frac{\pi}{4}$

2) b)

$\arg\left(1 + \sqrt{3}\mathrm{i}\right) = \arctan\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$

2) c)

$\arg(-2\mathrm{i}) = -\frac{\pi}{2}$

2) d)

$\arg(-2 - 2\mathrm{i}) = -\frac{\pi}{2} - \arctan\left(\frac{2}{2}\right) = -\frac{3\pi}{4}$

2) e)

$\arg(-3) = \pi$

3) a)

$3\mathrm{i}$

3) b)

$\frac{1}{8}+\frac{\sqrt{3}}{8}\mathrm{i}$

3) c)

$-\sqrt{3}+\mathrm{i}$

3) d)

$\frac{7\sqrt{3}}{2}-\frac{7}{2}\mathrm{i}$

3) e)

$-6-6\sqrt{3}\mathrm{i}$

3) f)

$\frac{1}{4}-\frac{1}{4}\mathrm{i}$

4) a)

$z=2\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)=1+\sqrt{3}\mathrm{i}$

4) b)

$z=1\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)=\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}\mathrm{i}$

4) c)

$z=4\left(\cos\left(-\frac{2\pi}{3}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{3}\right)\right)=-2-2\sqrt{3}\mathrm{i}$

5) a)

$|3+3\mathrm{i}|=\sqrt{3^2+3^2}=\sqrt{9+9}=\sqrt{18}$

$\arg(3+3\mathrm{i})=\arctan\left(\frac{3}{3}\right)=\frac{\pi}{4}$

$\therefore 3+3\mathrm{i}=\sqrt{18}\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)=3\sqrt{2}\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)$

5) b)

$|2\mathrm{i}| = \sqrt{0^2 + 2^2} = \sqrt{0 + 4} = 2$

$\arg(2\mathrm{i}) = \frac{\pi}{2}$

$\therefore 2\mathrm{i} = 2\left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)$

5) c)

$|-4| = \sqrt{0^2 + (-4)^2} = \sqrt{0 + 16} = 4$

$\arg(-4) = \pi$

$\therefore -4 = 4(\cos\pi + \mathrm{i}\sin\pi)$

5) d)

$\left|-1 + \sqrt{3}\mathrm{i}\right| = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{1 + 3} = 2$

$\arg\left(-1 + \sqrt{3}\mathrm{i}\right) = \frac{\pi}{2} + \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{2\pi}{3}$

$\therefore -1 + \sqrt{3}\mathrm{i} = 2\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right)$

5) e)

$|2 - 2\mathrm{i}| = \sqrt{(2)^2 + (-2)^2} = \sqrt{4 + 4} = 2\sqrt{2}$

$\arg(2 - 2\mathrm{i}) = -\arctan\left(\frac{2}{2}\right) = -\frac{\pi}{4}$

$\therefore 2 - 2\mathrm{i} = 2\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{\pi}{4}\right)\right)$

5) f)

$\left|-\sqrt{3} - \mathrm{i}\right| = \sqrt{\left(-\sqrt{3}\right)^2 + (-1)^2} = \sqrt{3 + 1} = 2$

$\arg\left(-\sqrt{3} - \mathrm{i}\right) = -\frac{\pi}{2} - \arctan\left(\frac{\sqrt{3}}{1}\right) = -\frac{5\pi}{6}$

$\therefore -\sqrt{3} - \mathrm{i} = 2\left(\cos\left(-\frac{5\pi}{6}\right) + \mathrm{i}\sin\left(-\frac{5\pi}{6}\right)\right)$

5) g)

$|-7 + 7\mathrm{i}| = \sqrt{(-7)^2 + 7^2} = \sqrt{49 + 49} = \sqrt{98} = 7\sqrt{2}$

$\arg(-7 + 7\mathrm{i}) = \frac{\pi}{2} + \arctan\left(\frac{7}{7}\right) = \frac{3\pi}{4}$

$\therefore -7 + 7\mathrm{i} = 7\sqrt{2}\left(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\right)$

5) h)

$$|3-\sqrt{3}\mathrm{i}| = \sqrt{3^2+\left(-\sqrt{3}\right)^2} = \sqrt{9+3} = \sqrt{12} = 2\sqrt{3}$$

$$\arg\left(3-\sqrt{3}\mathrm{i}\right) = -\arctan\left(\frac{\sqrt{3}}{3}\right) = -\frac{\pi}{6}$$

$$\therefore 3-\sqrt{3}\mathrm{i} = 2\sqrt{3}\left(\cos\left(-\frac{\pi}{6}\right)+\mathrm{i}\sin\left(-\frac{\pi}{6}\right)\right)$$

6) a)

$$|1+5\mathrm{i}| = \sqrt{1^2+5^2} = \sqrt{1+25} = \sqrt{26}$$

$$\arg(1+5\mathrm{i}) = \arctan\left(\frac{5}{1}\right) = 1.37$$

$$\therefore 1+5\mathrm{i} = \sqrt{26}(\cos 1.37+\mathrm{i}\sin 1.37)$$

6) b)

$$|-2+5\mathrm{i}| = \sqrt{(-2)^2+5^2} = \sqrt{4+25} = \sqrt{29}$$

$$\arg(-2+5\mathrm{i}) = \pi-\arctan\left(\frac{5}{2}\right) = 1.95$$

$$\therefore -2+5\mathrm{i} = \sqrt{29}(\cos 1.95+\mathrm{i}\sin 1.95)$$

6) c)

$$|4-\mathrm{i}| = \sqrt{4^2+(-1)^2} = \sqrt{16+1} = \sqrt{17}$$

$$\arg(4-\mathrm{i}) = -\arctan\left(\frac{1}{4}\right) = -0.245$$

$$\therefore 4-\mathrm{i} = \sqrt{17}(\cos(-0.245)+\mathrm{i}\sin(-0.245))$$

6) d)

$$|-3-7\mathrm{i}| = \sqrt{(-3)^2+(-7)^2} = \sqrt{9+49} = \sqrt{58}$$

$$\arg(-3-7\mathrm{i}) = -\pi+\arctan\left(\frac{7}{3}\right) = -1.98$$

$$\therefore -3-7\mathrm{i} = \sqrt{58}(\cos(-1.98)+\mathrm{i}\sin(-1.98))$$

6) e)

$$\left|\sqrt{2}+6\mathrm{i}\right| = \sqrt{\left(\sqrt{2}\right)^2+6^2} = \sqrt{2+36} = \sqrt{38}$$

$$\arg\left(\sqrt{2}+6\mathrm{i}\right) = \arctan\left(\frac{6}{\sqrt{2}}\right) = 1.34$$

$$\therefore \sqrt{2}+6\mathrm{i} = \sqrt{38}(\cos 1.34+\mathrm{i}\sin 1.34)$$

6) f)

$$\left|-1+\frac{3}{4}\mathrm{i}\right| = \sqrt{(-1)^2+\frac{3}{4}^2} = \sqrt{1+\frac{9}{16}} = \frac{5}{4}$$

$$\arg\left(-1+\frac{3}{4}\mathrm{i}\right) = \pi - \arctan\left(\frac{3}{4}\right) = 2.50$$

$$\therefore -1+\frac{3}{4}\mathrm{i} = \frac{5}{4}(\cos 2.50 + \mathrm{i}\sin 2.50)$$

6) g)

$$\left|2-\sqrt{3}\mathrm{i}\right| = \sqrt{2^2+\left(-\sqrt{3}\right)^2} = \sqrt{4+3} = \sqrt{7}$$

$$\arg\left(2-\sqrt{3}\mathrm{i}\right) = -\arctan\left(\frac{\sqrt{3}}{2}\right) = -0.714$$

$$\therefore 2-\sqrt{3}\mathrm{i} = \sqrt{7}(\cos(-0.714)+\mathrm{i}\sin(-0.714))$$

6) h)

$$\left|-1-\frac{1}{2}\mathrm{i}\right| = \sqrt{(-1)^2+\left(-\frac{1}{2}\right)^2} = \sqrt{1+\frac{1}{4}} = \frac{\sqrt{5}}{2}$$

$$\arg\left(-1-\frac{1}{2}\mathrm{i}\right) = -\pi + \arctan\left(\frac{1}{2}\right) = -2.68$$

$$\therefore -1-\frac{1}{2}\mathrm{i} = \frac{\sqrt{5}}{2}(\cos(-2.68)+\mathrm{i}\sin(-2.68))$$

7) a)

The mistake is with the angle; the angle of $\frac{\pi}{6}$ gives a complex number with positive real and imaginary parts, but the number has both a negative real and imaginary part.

7) b)

$$\arg\left(-\sqrt{3}-\mathrm{i}\right) = -\pi + \arctan\left(\frac{1}{\sqrt{3}}\right) = -\frac{5\pi}{6}$$

$$\therefore -\sqrt{3}-\mathrm{i} = 2\left(\cos\left(-\frac{5\pi}{6}\right)+\mathrm{i}\sin\left(-\frac{5\pi}{6}\right)\right)$$

8) a)

$$\frac{5\pi}{2} = 2\pi + \frac{\pi}{2}$$

$$\therefore 4\left(\cos\frac{\pi}{2}+\mathrm{i}\sin\frac{\pi}{2}\right)$$

8) b)

$-\sin\left(\frac{\pi}{3}\right) = \sin\left(-\frac{\pi}{3}\right)$ and $\cos\left(\frac{\pi}{3}\right) = \cos\left(-\frac{\pi}{3}\right)$

$\therefore 2\left(\cos\left(-\frac{\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{\pi}{3}\right)\right)$

8) c)

$\frac{10\pi}{3} = 2\pi + \frac{4\pi}{3} \therefore -\frac{2\pi}{3}$

$-\sin\left(-\frac{2\pi}{3}\right) = \sin\left(\frac{2\pi}{3}\right)$ and $\cos\left(-\frac{2\pi}{3}\right) = \cos\left(\frac{2\pi}{3}\right)$

$\therefore 3\left(\cos\left(\frac{2\pi}{3}\right) + \mathrm{i}\sin\left(\frac{2\pi}{3}\right)\right)$

8) d)

$\frac{10\pi}{3} = 2\pi + \frac{4\pi}{3}$ giving $-5\left(\cos\left(\frac{4\pi}{3}\right) + \mathrm{i}\sin\left(\frac{4\pi}{3}\right)\right)$

$r = -5$ becoming $r = 5$ gives: $5\left(\cos\left(\frac{\pi}{3}\right) + \mathrm{i}\sin\left(\frac{\pi}{3}\right)\right)$

9) a)

$\frac{1}{1-3\mathrm{i}} \times \frac{1+3\mathrm{i}}{1+3\mathrm{i}} = \frac{1}{10} + \frac{3}{10}\mathrm{i}$

$\left|\frac{1}{10} + \frac{3}{10}\mathrm{i}\right| = \sqrt{\left(\frac{1}{10}\right)^2 + \left(\frac{3}{10}\right)^2} = \frac{1}{\sqrt{10}}$ and $\arg z = \arctan(3) = 1.25$

$z = \frac{1}{\sqrt{10}}(\cos 1.25 + \mathrm{i}\sin 1.25)$

Or $z = \frac{\sqrt{10}}{10}(\cos 1.25 + \mathrm{i}\sin 1.25)$

9) b)

$\frac{1}{1+3\mathrm{i}} \times \frac{1-3\mathrm{i}}{1-3\mathrm{i}} = \frac{1}{10} - \frac{3}{10}\mathrm{i}$

$\left|\frac{1}{10} - \frac{3}{10}\mathrm{i}\right| = \sqrt{\left(\frac{1}{10}\right)^2 + \left(-\frac{3}{10}\right)^2} = \frac{1}{\sqrt{10}}$ and $\arg z = \arctan(-3) = -1.25$

$z = \frac{1}{\sqrt{10}}(\cos(-1.25) + \mathrm{i}\sin(-1.25))$

or $z = \frac{\sqrt{10}}{10}(\cos(-1.25) + \mathrm{i}\sin(-1.25))$

9) c)

$\frac{5}{2-3i} \times \frac{2+3i}{2+3i} = \frac{10+15i}{4+9} = \frac{10}{13} + \frac{15}{13}i$

$\left|\frac{10}{13} + \frac{15}{13}i\right| = \sqrt{\left(\frac{10}{13}\right)^2 + \left(\frac{15}{13}\right)^2} = \frac{5\sqrt{13}}{13}$ and $\arg z = \arctan\left(\frac{15}{10}\right) = 0.983$

$z = \frac{5\sqrt{13}}{13}\ (\cos(0.983) + i\sin(0.983))$

9) d)

$\frac{-7+i}{7-2i} \times \frac{7+2i}{7+2i} = \frac{-51-7i}{49+4} = -\frac{51}{53} - \frac{7}{53}i$

$\left|-\frac{51}{53} - \frac{7}{53}i\right| = \sqrt{\left(-\frac{51}{53}\right)^2 + \left(-\frac{7}{53}\right)^2} = \frac{5\sqrt{106}}{53}$ and $\arg z = -\pi + \arctan\left(\frac{7}{51}\right) = -3.01$

$z = \frac{5\sqrt{106}}{53}\ (\cos(-3.01) + i\sin(-3.01))$

10)

$3\left(\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right)$

11) a)

$|4 - 5i| = \sqrt{4^2 + (-5)^2} = \sqrt{41}$ and $\arg z = -\arctan\left(\frac{5}{4}\right) = -0.896$

$z = \sqrt{41}\ (\cos(-0.896) + i\sin(-0.896))$

11) b)

$z^* = \sqrt{41}\ (\cos(0.896) + i\sin(0.896))$

12) a)

$z^2 - 2z + 2 = 0$ gives $(z-1)^2 + 1 = 0 \Rightarrow z = 1 \pm i$

$|1 \pm i| = \sqrt{(1)^2 + (\pm 1)^2} = \sqrt{2}$

$\arg(1 + i) = \arctan(1) = \frac{\pi}{4}$

$\arg(1 - i) = -\arctan(1) = -\frac{\pi}{4}$

$z = \sqrt{2}\left(\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right)$ or $z = \sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right)$

12) b)

$9z^2 - 3z + 1 = 0$ gives $9\left(z - \frac{1}{6}\right)^2 + \frac{3}{4} = 0 \Rightarrow z = \frac{1}{6} \pm \frac{\sqrt{3}}{6}\mathrm{i}$

$\left|\frac{1}{6} \pm \frac{\sqrt{3}}{6}\mathrm{i}\right| = \sqrt{\left(\frac{1}{6}\right)^2 + \left(\pm\frac{\sqrt{3}}{6}\right)^2} = \frac{1}{3}$

$\arg\left(\frac{1}{6} + \frac{\sqrt{3}}{6}\mathrm{i}\right) = \arctan\left(\sqrt{3}\right) = \frac{\pi}{3}$

$\arg\left(\frac{1}{6} - \frac{\sqrt{3}}{6}\mathrm{i}\right) = -\arctan\left(\sqrt{3}\right) = -\frac{\pi}{3}$

$z = \frac{1}{3}\left(\cos\left(\frac{\pi}{3}\right) + \mathrm{i}\sin\left(\frac{\pi}{3}\right)\right)$ or $z = \frac{1}{3}\left(\cos\left(-\frac{\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{\pi}{3}\right)\right)$

12) c)

$z^2 + 8z + 64 = 0$ gives $(z+4)^2 + 48 = 0 \Rightarrow z = -4 \pm 4\sqrt{3}\mathrm{i}$

$\left|-4 \pm 4\sqrt{3}\mathrm{i}\right| = \sqrt{(-4)^2 + \left(\pm 4\sqrt{3}\right)^2} = 8$

$\arg\left(-4 + 4\sqrt{3}\mathrm{i}\right) = \pi - \arctan\left(\frac{4\sqrt{3}}{4}\right) = \frac{2\pi}{3}$

$\arg\left(-4 - 4\sqrt{3}\mathrm{i}\right) = -\pi + \arctan\left(\frac{4\sqrt{3}}{4}\right) = -\frac{2\pi}{3}$

$z = 8\left(\cos\left(\frac{2\pi}{3}\right) + \mathrm{i}\sin\left(\frac{2\pi}{3}\right)\right)$ or $z = 8\left(\cos\left(-\frac{2\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{3}\right)\right)$

12) d)

$k^2z^2 - 2kz + 2 = 0$ gives $(kz-1)^2 + 1 = 0 \Rightarrow z = \frac{1}{k} \pm \frac{1}{k}\mathrm{i}$

$\left|\frac{1}{k} \pm \frac{1}{k}\mathrm{i}\right| = \sqrt{\left(\frac{1}{k}\right)^2 + \left(\pm\frac{1}{k}\right)^2} = \frac{\sqrt{2}}{k}$

$\arg\left(\frac{2}{k} + \frac{2}{k}\mathrm{i}\right) = \arctan(1) = \frac{\pi}{4}$

$\arg\left(\frac{2}{k} - \frac{2}{k}\mathrm{i}\right) = -\arctan(1) = -\frac{\pi}{4}$

$z = \frac{\sqrt{2}}{k}\left(\cos\left(\frac{\pi}{4}\right) + \mathrm{i}\sin\left(\frac{\pi}{4}\right)\right)$ or $z = \frac{\sqrt{2}}{k}\left(\cos\left(-\frac{\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{\pi}{4}\right)\right)$

13) a)

$|2 + 3\text{i}| = \sqrt{2^2 + 3^2} = \sqrt{13}$

$\arg(2 + 3\text{i}) = \arctan\left(\frac{3}{2}\right) = 0.983$

$\therefore z_1 = 2 + 3\text{i} = \sqrt{13}(\cos 0.983 + \text{i}\sin 0.983)$

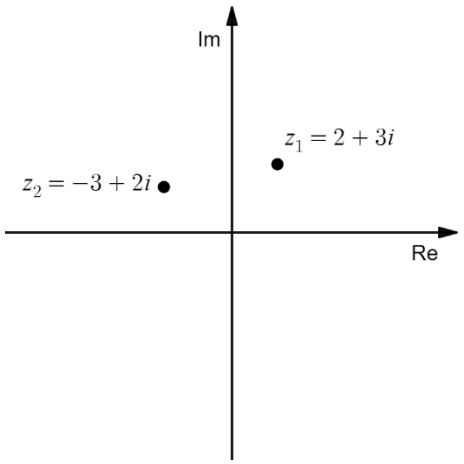
$z_2 = -3 + 2\text{i}$

$|-3 + 2\text{i}| = \sqrt{(-3)^2 + 2^2} = \sqrt{13}$

$\arg(-3 + 2\text{i}) = \pi - \arctan\left(\frac{2}{3}\right) = 2.55$

$\therefore z_1 = \sqrt{13}(\cos 0.983 + \text{i}\sin 0.983)$ and $z_2 = -3 + 2\text{i} = \sqrt{13}(\cos 2.55 + \text{i}\sin 2.55)$

13) b)



13) c)

Rotation $\frac{\pi}{2}$ anti-clockwise about the origin

14) a)

$|1 + 2\text{i}| = \sqrt{1^2 + 2^2} = \sqrt{5}$

$\arg(1 + 2\text{i}) = \arctan\left(\frac{2}{1}\right) = 1.11$

$\therefore z_1 = 1 + 2\text{i} = \sqrt{5}(\cos 1.11 + \text{i}\sin 1.11)$

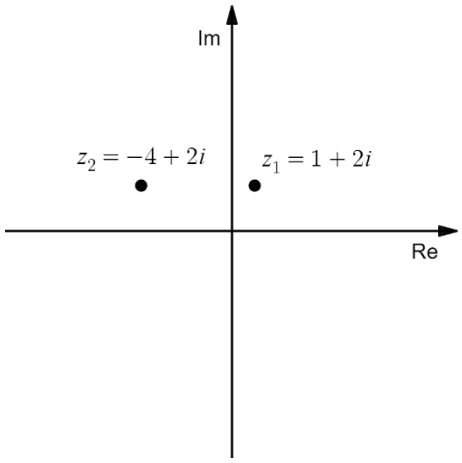
$z_2 = 2\text{i}(1 + 2\text{i}) = -4 + 2\text{i}$

$|-4 + 2\text{i}| = \sqrt{(-4)^2 + 2^2} = 2\sqrt{5}$

$\arg(-4 + 2\text{i}) = \pi - \arctan\left(\frac{2}{4}\right) = 2.68$

$\therefore z_1 = \sqrt{5}(\cos 1.11 + \text{i}\sin 1.11)$ and $z_2 = 2\sqrt{5}(\cos 2.68 + \text{i}\sin 2.68)$

14) b)



14) c)

$\pi - \arctan\left(\frac{2}{4}\right) - \arctan\left(\frac{2}{1}\right) = \frac{\pi}{2}$ and $2\sqrt{5} \div \sqrt{5} = 2$

Rotation $\frac{\pi}{2}$ anti-clockwise about the origin

Enlargement scale factor 2

15) a)

$|3 + 2\mathrm{i}| = \sqrt{3^2 + 2^2} = \sqrt{13}$

$\arg(3 + 2\mathrm{i}) = \arctan\left(\frac{2}{3}\right) = 0.588$

$\therefore z_1 = 3 + 2\mathrm{i} = \sqrt{13}(\cos 0.588 + \mathrm{i} \sin 0.588)$

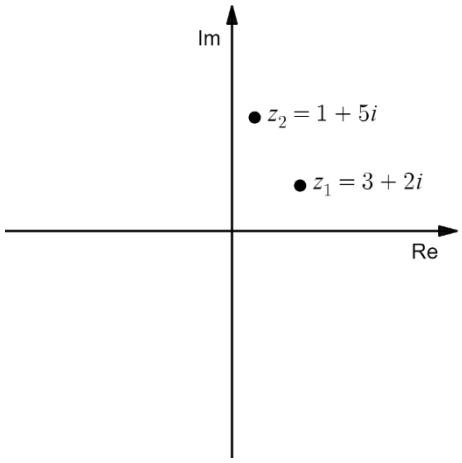
$z_2 = (1 + \mathrm{i})(3 + 2\mathrm{i}) = 1 + 5\mathrm{i}$

$|1 + 5\mathrm{i}| = \sqrt{(1)^2 + 5^2} = \sqrt{26}$

$\arg(1 + 5\mathrm{i}) = \arctan\left(\frac{5}{1}\right) = 1.37$

$\therefore z_1 = \sqrt{13}(\cos 0.588 + \mathrm{i} \sin 0.588)$ and $z_2 = \sqrt{26}(\cos 1.37 + \mathrm{i} \sin 1.37)$

15) b)



15) c)

$\arctan\left(\frac{5}{1}\right) - \arctan\left(\frac{2}{3}\right) = \frac{\pi}{4}$ and $\sqrt{26} \div \sqrt{13} = \sqrt{2}$

Rotation $\frac{\pi}{4}$ anti-clockwise about the origin

Enlargement scale factor $\sqrt{2}$

16) a)

$3 + 2\text{i} + 1 - 5\text{i} = 4 - 3\text{i} \therefore |4 - 3\text{i}| = \sqrt{4^2 + (-3)^2} = 5$

$|3 + 2\text{i}| = \sqrt{3^2 + 2^2} = \sqrt{13}$

$|1 - 5\text{i}| = \sqrt{1^2 + (-5)^2} = \sqrt{26}$

$\sqrt{13} + \sqrt{26} = 8.70$ (3 SF) and $5 < 8.70$

So $|z_1 + z_2| \leq |z_1| + |z_2|$

16) b)

By considering complex numbers as vectors: z_1, z_2 and $z_1 + z_2$ form a triangle

For any triangle the length of one side is always less than or equal to the sum of the lengths of the other two sides.

$\therefore |z_1 + z_2| \leq |z_1| + |z_2|$

17) a)

$\arg(a - b\text{i}) = -\frac{\pi}{4}$

17) b)

$\arg(-a-b\mathrm{i}) = -\frac{3\pi}{4}$

17) c)

$\arg(b+a\mathrm{i}) = \frac{\pi}{4}$

17) d)

$\arg(b-a\mathrm{i}) = -\frac{\pi}{4}$

18) a)

$\arg(a-b\mathrm{i}) = \frac{\pi}{3}$

18) b)

$\arg(-a-b\mathrm{i}) = \frac{2\pi}{3}$

18) c)

$\arg(b+a\mathrm{i}) = \frac{5\pi}{6}$

18) d)

$\arg(b-a\mathrm{i}) = -\frac{5\pi}{6}$

19) a)

$\theta = \arctan\left(\frac{2}{1}\right) - \arctan\left(\frac{3}{4}\right) = 0.464$ (3 SF)

Alternative Method:

$\arg\left(\frac{z}{w}\right) = \arg\left(\frac{4+3\mathrm{i}}{1+2\mathrm{i}}\right) = \arg(2-\mathrm{i}) = -\arctan\left(\frac{1}{2}\right) = -0.464$ (3 SF) $\therefore 0.464$ (3 SF)

19) b)

$\theta = \pi - \arctan\left(\frac{6}{3}\right) - \arctan\left(\frac{2}{2}\right) = 1.25$ (3 SF)

Alternative Method:

$\arg\left(\frac{z}{w}\right) = \arg\left(\frac{2+2\mathrm{i}}{-3+6\mathrm{i}}\right) = \arg\left(\frac{2}{15} - \frac{6}{15}\mathrm{i}\right) = -\arctan(3) = -1.24$ (3 SF) $\therefore 1.24$ (3 SF)

19) c)

$\theta = \frac{\pi}{2} + \arctan\left(\frac{3}{4}\right) + \arctan\left(\frac{1}{1}\right) = 3.00$ (3 SF)

Alternative Method:

$\arg\left(\frac{z}{w}\right) = \arg\left(\frac{-4+3\mathrm{i}}{1-\mathrm{i}}\right) = \arg\left(-\frac{7}{2} - \frac{1}{2}\mathrm{i}\right) = -\pi + \arctan\left(\frac{1}{7}\right) = -3.00$ (3 SF) $\therefore 3.00$ (3 SF)

20) a)

$|z_1| = \sqrt{2^2 + 1^2} = \sqrt{5}$ and $\arg z_1 = \arctan\left(\frac{1}{2}\right)$

$|z_2| = \sqrt{1^2 + 3^2} = \sqrt{10}$ and $\arg z_2 = \arctan(3)$

$\arctan(3) - \arctan\left(\frac{1}{2}\right) = \frac{\pi}{4}$

Area:

$\frac{1}{2} \times \sqrt{5} \times \sqrt{10} \sin\left(\frac{\pi}{4}\right) = \frac{1}{2} \times \sqrt{5} \times \sqrt{10} \times \frac{\sqrt{2}}{2} = \frac{5}{2}$ units^2

Alternative Method:

$|z_1| = \sqrt{2^2 + 1^2} = \sqrt{5}$

$|z_2| = \sqrt{1^2 + 3^2} = \sqrt{10}$

$\arg\left(\frac{z_1}{z_2}\right) = \arg\left(\frac{2+\mathrm{i}}{1+3\mathrm{i}}\right) = \arg\left(\frac{1}{2} - \frac{1}{2}\mathrm{i}\right) = -\arctan 1 = -\frac{\pi}{4}$ (3 SF) $\therefore \frac{\pi}{4}$ (3 SF)

Area:

$\frac{1}{2} \times \sqrt{5} \times \sqrt{10} \sin\left(\frac{\pi}{4}\right) = \frac{1}{2} \times \sqrt{5} \times \sqrt{10} \times \frac{\sqrt{2}}{2} = \frac{5}{2}$ units^2

20) b)

$|z_1| = \sqrt{1^2 + 2^2} = \sqrt{5}$ and $\arg z_1 = \arctan(2)$

$|z_2| = \sqrt{(-2)^2 + 4^2} = 2\sqrt{5}$ and $\arg z_2 = \pi - \arctan\left(\frac{4}{2}\right)$

$\pi - \arctan\left(\frac{4}{2}\right) - \arctan(2) = 0.927\ldots$

Area:

$\frac{1}{2} \times \sqrt{5} \times 2\sqrt{5} \sin(0.927\ldots) = \frac{1}{2} \times \sqrt{5} \times 2\sqrt{5} \times \frac{4}{5} = 4$ units^2

Alternative Method:

$|z_1| = \sqrt{1^2 + 2^2} = \sqrt{5}$

$|z_2| = \sqrt{(-2)^2 + 4^2} = 2\sqrt{5}$

$\arg\left(\frac{z_1}{z_2}\right) = \arg\left(\frac{1+2\mathrm{i}}{-2+4\mathrm{i}}\right) = \arg\left(\frac{3}{10} - \frac{2}{5}\mathrm{i}\right) = -\arctan\left(\frac{4}{3}\right) = -0.927$ (3 SF) $\therefore 0.927$ (3 SF)

Area:

$\frac{1}{2} \times \sqrt{5} \times 2\sqrt{5} \sin(0.927 \ldots) = \frac{1}{2} \times \sqrt{5} \times 2\sqrt{5} \times \frac{4}{5} = 4$ units^2

20) c)

$|z_1| = \sqrt{(-5)^2 + 3^2} = \sqrt{34}$ and $|z_2| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$

$\arctan\left(\frac{3}{5}\right) + \frac{\pi}{2} + \arctan\left(\frac{2}{3}\right) = 2.699 \ldots$

Area:

$\frac{1}{2} \times \sqrt{34} \times \sqrt{13} \sin(2.699 \ldots) = \frac{1}{2} \times \sqrt{34} \times \sqrt{13} \times 0.428 \ldots = \frac{9}{2}$ units^2

Alternative Method:

$|z_1| = \sqrt{(-5)^2 + 3^2} = \sqrt{34}$

$|z_2| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$

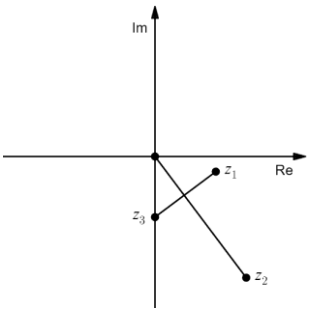
$\arg\left(\frac{z_1}{z_2}\right) = \arg\left(\frac{-5+3\mathrm{i}}{2-3\mathrm{i}}\right) = \arg\left(-\frac{19}{13} - \frac{9}{13}\mathrm{i}\right) = -\pi + \arctan\left(\frac{9}{19}\right) = -2.699 \ldots$ (3 SF)

$\therefore 2.699 \ldots$ (3 SF)

Area:

$\frac{1}{2} \times \sqrt{34} \times \sqrt{13} \sin(2.699 \ldots) = \frac{1}{2} \times \sqrt{34} \times \sqrt{13} \times 0.428 \ldots = \frac{9}{2}$ units^2

21)



Converting to cartesian:

$m_1 = -\frac{8}{6} = -\frac{4}{3}$ and $m_2 = \frac{-1--4}{4+0} = \frac{3}{4}$ so the diagonals are perpendicular so it is a kite.

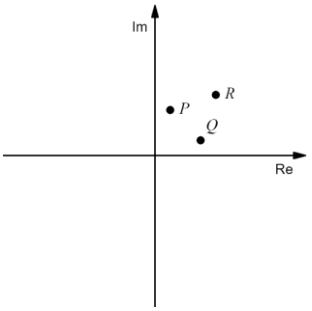
$z_3 - z_1 = -4 - 3\mathrm{i}$ so $|z_3 - z_1| = \sqrt{(-4)^2 + (-3)^2} = 5$

$|z_2| = \sqrt{6^2 + 8^2} = 10$

$\frac{1}{2} \times 5 \times 10 = 25$

Therefore it has an area of 25 units^2

22) a)



22) b)

Let $\arg(z_1) = \theta_1$ and $\arg(z_2) = \theta_2$

The diagonal of the rhombus bisects the angle $\angle POQ$

So $\arg(z_1 + z_2) = \frac{1}{2}(\theta_1 - \theta_2) + \theta_2$

$= \frac{1}{2}\theta_1 + \frac{1}{2}\theta_2$

$= \frac{1}{2}\arg(z_1) + \frac{1}{2}\arg(z_2)$

$= \frac{1}{2}(\arg z_1 + \arg z_2)$ as req.

23) a)

$3 \times 5 = 15, \frac{\pi}{4} + \frac{\pi}{3} = \frac{7\pi}{12}$

$zw = 15\left(\cos\frac{7\pi}{12} + \mathrm{i}\sin\frac{7\pi}{12}\right)$

23) b)

$3 \div 5 = \frac{3}{5}, \frac{\pi}{4} - \frac{\pi}{3} = -\frac{\pi}{12}$

$\frac{z}{w} = \frac{3}{5}\left(\cos\left(-\frac{\pi}{12}\right) + \mathrm{i}\sin\left(-\frac{\pi}{12}\right)\right)$

23) c)

$5 \div 3 = \frac{5}{3}, \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$

$\frac{w}{z} = \frac{5}{3}\left(\cos\frac{\pi}{12} + \mathrm{i}\sin\frac{\pi}{12}\right)$

23) d)

$3 \times 3 = 9, \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$

$z^2 = 9\left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)$

23) e)

$5^3 = 125, \frac{\pi}{3} + \frac{\pi}{3} + \frac{\pi}{3} = \pi$

$w^3 = 125(\cos\pi + \mathrm{i}\sin\pi)$

24) a)

$2 \times 4 = 8, 1.4 + 0.7 = 2.1$

$zw = 8(\cos 2.1 + \mathrm{i}\sin 2.1)$

24) b)

$2 \div 4 = \frac{1}{2}, 1.4 - 0.7 = 0.7$

$\frac{z}{w} = \frac{1}{2}(\cos 0.7 + \mathrm{i}\sin 0.7)$

24) c)

$4 \div 2 = 2, 0.7 - 1.4 = -0.7$

$\frac{w}{z} = 2(\cos(-0.7) + \mathrm{i}\sin(-0.7))$

24) d)

$4^2 = 16, 0.7 + 0.7 = 1.4$

$w^2 = 16(\cos(1.4) + \mathrm{i}\sin(1.4))$

24) e)

$4^2 \times 2 = 32, 0.7 + 0.7 + 1.4 = 2.8$

$w^2 z = 32(\cos(2.8) + \mathrm{i}\sin(2.8))$

24) f)

$2^2 \div 4^2 = \frac{1}{4}, 1.4 + 1.4 - 0.7 - 0.7 = 1.4.$

$\frac{z^2}{w^2} = \frac{1}{4}(\cos 1.4 + \mathrm{i}\sin 1.4)$

25) a)

$7 \times 10 = 70, \frac{\pi}{6} + \left(-\frac{\pi}{3}\right) = -\frac{\pi}{6}$

$zw = 70\left(\cos\left(-\frac{\pi}{6}\right) + \mathrm{i}\sin\left(-\frac{\pi}{6}\right)\right)$

25) b)

$7 \div 10 = \frac{7}{10}, \frac{\pi}{6} - \left(-\frac{\pi}{3}\right) = \frac{\pi}{2}$

$\frac{z}{w} = \frac{7}{10}\left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)$

25) c)

$10 \div 7 = \frac{10}{7}, \left(-\frac{\pi}{3}\right) - \frac{\pi}{6} = -\frac{\pi}{2}$

$\frac{w}{z} = \frac{10}{7}\left(\cos\left(-\frac{\pi}{2}\right) + \mathrm{i}\sin\left(-\frac{\pi}{2}\right)\right)$

25) d)

$10^2 = 100, \left(-\frac{\pi}{3}\right) + \left(-\frac{\pi}{3}\right) = -\frac{2\pi}{3}$

$2w^2 = 200\left(\cos\left(-\frac{2\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{3}\right)\right)$

26)

$|1 + \mathrm{i}| = \sqrt{1^2 + 1^2} = \sqrt{2}$ and $\arg z = \arctan(1) = \frac{\pi}{4}$

$z = \sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)$

$\left|\sqrt{3} + \mathrm{i}\right| = \sqrt{\left(\sqrt{3}\right)^2 + 1^2} = 2$ and $\arg z = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

$w = 2\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)$

$\sqrt{2} \times 2^2 = 4\sqrt{2}, \frac{\pi}{4} + \frac{\pi}{6} + \frac{\pi}{6} = \frac{7\pi}{12}$

$zw^2 = 4\sqrt{2}\left(\cos\frac{7\pi}{12} + \mathrm{i}\sin\frac{7\pi}{12}\right)$

27) a)

$2 \times 5 = 10, \frac{\pi}{4} + \frac{4\pi}{5} = \frac{21\pi}{20}$

But $\frac{21\pi}{20} > \pi$ so $-\frac{19\pi}{20}$

$zw = 10\left(\cos\left(-\frac{19\pi}{20}\right) + \mathrm{i}\sin\left(-\frac{19\pi}{20}\right)\right)$

27) b)

$2 \div 5 = \frac{2}{5}, \frac{\pi}{4} - \frac{4\pi}{5} = -\frac{11\pi}{20}$

$\frac{z}{w} = \frac{2}{5}\left(\cos\left(-\frac{11\pi}{20}\right) + \mathrm{i}\sin\left(-\frac{11\pi}{20}\right)\right)$

27) c)

$5^3 = 125, \frac{4\pi}{5} + \frac{4\pi}{5} + \frac{4\pi}{5} = \frac{12\pi}{5}$

But $\frac{12\pi}{5} > \pi$ so $\frac{2\pi}{5}$

$w^3 = 125\left(\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}\right)$

27) d)

$(1+\mathrm{i})z = (1+\mathrm{i}) \times 2\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)$

Since $1 + \mathrm{i} = \sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)$ then:

$(1+\mathrm{i})z = 2\sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right) = 2\sqrt{2}\left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)$

28) a)

$2 \times \frac{1}{2} = 1, -\frac{5\pi}{6} + \frac{3\pi}{4} = -\frac{\pi}{12}$

$zw = 1\left(\cos\left(-\frac{\pi}{12}\right) + \mathrm{i}\sin\left(-\frac{\pi}{12}\right)\right)$

28) b)

$2 \div \frac{1}{2} = 4, -\frac{5\pi}{6} - \frac{3\pi}{4} = -\frac{19\pi}{12}$

But $-\frac{19\pi}{12} < -\pi$ so $\frac{5\pi}{12}$

$\frac{z}{w} = 4\left(\cos\frac{5\pi}{12} + \mathrm{i}\sin\frac{5\pi}{12}\right)$

28) c)

$2^2 \times \left(\frac{1}{2}\right)^2 = 1, \left(-\frac{5\pi}{6}\right) + \left(-\frac{5\pi}{6}\right) + \frac{3\pi}{4} + \frac{3\pi}{4} = -\frac{\pi}{6}$

$(zw)^2 = 1\left(\cos\left(-\frac{\pi}{6}\right) + \mathrm{i}\sin\left(-\frac{\pi}{6}\right)\right)$

28) d)

$(1+\sqrt{3}\mathrm{i})w = (1+\sqrt{3}\mathrm{i}) \times \frac{1}{2}\left(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\right)$

Since $1+\sqrt{3}\mathrm{i} = 2\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)$ then:

$(1+\sqrt{3}\mathrm{i})w = \frac{1}{2}\left(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\right) \times 2\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)$

$= 1\left(\cos\frac{13\pi}{12} + \mathrm{i}\sin\frac{13\pi}{12}\right)$

$= 1\left(\cos\left(-\frac{11\pi}{12}\right) + \mathrm{i}\sin\left(-\frac{11\pi}{12}\right)\right)$

29) a)

$|z| = \sqrt{\left(\sqrt{3}\right)^2 + \left(\sqrt{3}\right)^2} = \sqrt{6}$ and $\arg z = \arctan(1) = \frac{\pi}{4}$

$z = \sqrt{6}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)$

29) b)

$|w| = \sqrt{0^2 + 4^2} = 4$ and $\arg w = \frac{\pi}{4}$

$w = 4\left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)$

29) c)

$\sqrt{6} \times 4^2 = 16\sqrt{6}$ and $\frac{\pi}{4} + \frac{\pi}{2} + \frac{\pi}{2} = \frac{5\pi}{4}$

But $\frac{5\pi}{4} > \pi$ so $-\frac{3\pi}{4}$

$zw^2 = 16\sqrt{6}\left(\cos\left(-\frac{3\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{3\pi}{4}\right)\right)$

30) a)

$z = 3\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right) = -\frac{3}{2} + \frac{3\sqrt{3}}{2}\mathrm{i}$

30) b)

$z^* = 3\left(\cos\left(-\frac{2\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{3}\right)\right) = -\frac{3}{2} - \frac{3\sqrt{3}}{2}\mathrm{i}$

30) c)

$z^2 = 9\left(\cos\frac{4\pi}{3} + \mathrm{i}\sin\frac{4\pi}{3}\right)$

$z^2 = 9\left(\cos\left(-\frac{2\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{3}\right)\right) = -\frac{9}{2} - \frac{9\sqrt{3}}{2}\mathrm{i}$

30) d)

$3 \div 3 = 1, \frac{2\pi}{3} - \left(-\frac{2\pi}{3}\right) = \frac{4\pi}{3}$

$\frac{z}{z^*} = 1\left(\cos\frac{4\pi}{3} + \mathrm{i}\sin\frac{4\pi}{3}\right) = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$

31) a)

$\arg(z^2) = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$

31) b)

$\arg(z^2) = -\frac{\pi}{3} - \frac{\pi}{3} = -\frac{2\pi}{3}$

32) a)

$\frac{z}{w} = \frac{2}{3}\left(\cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) + \mathrm{i}\sin\left(\frac{\pi}{3} - \frac{\pi}{4}\right)\right) = \frac{2}{3}\left(\cos\frac{\pi}{12} + \mathrm{i}\sin\frac{\pi}{12}\right) = 0.644 + 0.173\mathrm{i}$

32) b)

$|1 + \mathrm{i}| = \sqrt{1^2 + 1^2} = \sqrt{2}$

$\arg(1 + \mathrm{i}) = \arctan(1) = \frac{\pi}{4}$

$\frac{z}{w} = \frac{\sqrt{2}}{2}\left(\cos\left(\frac{\pi}{4} - \frac{\pi}{6}\right) + \mathrm{i}\sin\left(\frac{\pi}{4} - \frac{\pi}{6}\right)\right) = \frac{\sqrt{2}}{2}\left(\cos\frac{\pi}{12} + \mathrm{i}\sin\frac{\pi}{12}\right) = 0.683 + 0.183\mathrm{i}$

33) a)

$|5 - 2\mathrm{i}| = \sqrt{5^2 + (-2)^2} = \sqrt{29}$

$\frac{\sqrt{290}}{\sqrt{29}} = \sqrt{10} \therefore |w| = \sqrt{10}$

33) b)

$\arg(5 - 2\mathrm{i}) = -\arctan\left(-\frac{2}{5}\right)$

$\arg(w) = 0.8 - \arctan\left(-\frac{2}{5}\right) \therefore \arg(w) = 1.18$ (3 SF)

34) a)

$|z_1| = \sqrt{\left(-\frac{\sqrt{3}}{2}\right)^2 + \left(-\frac{3}{2}\right)^2} = \sqrt{3}$ and $\arg z = -\pi + \arctan\left(\frac{-\frac{3}{2}}{-\frac{\sqrt{3}}{2}}\right) = -\frac{2\pi}{3}$

$z_1 = \sqrt{3}\left(\cos\left(-\frac{2\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{3}\right)\right)$

34) b)

$|z_1 z_2| = \sqrt{30} = \sqrt{3}|z_2|$

so $|z_2| = \sqrt{10}$

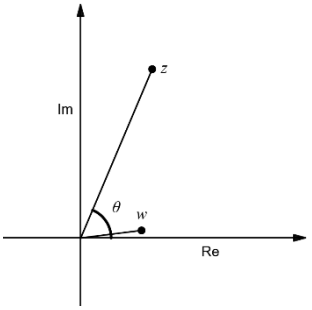
$\arg(z_1 z_2) = \frac{\pi}{4}$ so $-\frac{2\pi}{3} + \arg(z_2) = \frac{\pi}{4}$ so $\arg(z_2) = \frac{\pi}{4} + \frac{2\pi}{3} = \frac{11\pi}{12}$

$\therefore z_2 = \sqrt{10}\left(\cos\left(\frac{11\pi}{12}\right) + \mathrm{i}\sin\left(\frac{11\pi}{12}\right)\right)$

$= \left(-\frac{\sqrt{15}}{2} - \frac{\sqrt{5}}{2}\right) + \left(\frac{\sqrt{15}}{2} - \frac{\sqrt{5}}{2}\right)\mathrm{i}$

35)

Let $\arg z = \theta$



and so $\arg w = \theta - \frac{\pi}{3}$ since w has the smaller argument

$\arg\left(\frac{w}{z}\right) = \arg(w) - \arg(z) = \theta - \frac{\pi}{3} - \theta = -\frac{\pi}{3}$ and $\left|\frac{w}{z}\right| = \frac{|w|}{|z|} = \frac{|3z|}{|z|} = 3$

$\frac{w}{z} = 3\left(\cos\left(-\frac{\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{\pi}{3}\right)\right) = \frac{3}{2} - \frac{3\sqrt{3}}{2}\mathrm{i}$

36) a)

$|z| = \sqrt{(-k)^2 + k^2} = \sqrt{2k^2} = \sqrt{2}k$ and $\arg z = \pi - \arctan(1) = \frac{3\pi}{4}$

$z = \sqrt{2}k\left(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\right)$

36) b)

$zw = \sqrt{2}k \times 2\sqrt{2}(\cos\pi + \mathrm{i}\sin\pi)$

$-20 = 4k\cos\pi \Rightarrow k = 5$

37)

$\mathrm{i}z^3 = 2^3\mathrm{i}\left(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\right)$

$= 8\mathrm{i}\left(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\right)$

Since $\mathrm{i} = \cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}$ then:

$\mathrm{i}z^2 = 8\left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)\left(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\right)$

$= 8\left(\cos\frac{\pi}{2}\cos\frac{3\pi}{4} - \sin\frac{\pi}{2}\sin\frac{3\pi}{4} + \mathrm{i}\sin\frac{\pi}{2}\cos\frac{3\pi}{4} + \mathrm{i}\cos\frac{\pi}{2}\sin\frac{3\pi}{4}\right)$

$= 8\left(\cos\left(\frac{\pi}{2} + \frac{3\pi}{4}\right) + \mathrm{i}\sin\left(\frac{\pi}{2} + \frac{3\pi}{4}\right)\right)$

$= 8\left(\cos\frac{5\pi}{4} + \mathrm{i}\sin\frac{5\pi}{4}\right)$

$= 8\left(\cos\left(-\frac{3\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{3\pi}{4}\right)\right)$

38) a)

Let $z_1 = r_1(\cos\theta_1 + \mathrm{i}\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + \mathrm{i}\sin\theta_2)$

$z_1z_2 = r_1(\cos\theta_1 + \mathrm{i}\sin\theta_1) \times r_2(\cos\theta_2 + \mathrm{i}\sin\theta_2)$

$= r_1r_2(\cos\theta_1 + \mathrm{i}\sin\theta_1)(\cos\theta_2 + \mathrm{i}\sin\theta_2)$

$= r_1r_2(\cos\theta_1\cos\theta_2 + \mathrm{i}\cos\theta_1\sin\theta_2 + \mathrm{i}\sin\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2)$

$= r_1r_2\big((\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + \mathrm{i}(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)\big)$

$= r_1r_2(\cos(\theta_1 + \theta_2) + \mathrm{i}\sin(\theta_1 + \theta_2))$

So $|z_1z_2| = r_1r_2$ and $\arg(z_1z_2) = \theta_1 + \theta_2$

38) b)

Let $z_1 = r_1(\cos\theta_1 + \mathrm{i}\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + \mathrm{i}\sin\theta_2)$

$\frac{z_1}{z_2} = r_1(\cos\theta_1 + \mathrm{i}\sin\theta_1) \div r_2(\cos\theta_2 + \mathrm{i}\sin\theta_2)$

$= \frac{r_1}{r_2} \times \frac{(\cos\theta_1 + \mathrm{i}\sin\theta_1)}{(\cos\theta_2 + \mathrm{i}\sin\theta_2)}$

$= \frac{r_1}{r_2} \times \frac{(\cos\theta_1 + \mathrm{i}\sin\theta_1)}{(\cos\theta_2 + \mathrm{i}\sin\theta_2)} \times \frac{(\cos\theta_2 - \mathrm{i}\sin\theta_2)}{(\cos\theta_2 - \mathrm{i}\sin\theta_2)}$

$= \frac{r_1}{r_2} \times \frac{(\cos\theta_1 + \mathrm{i}\sin\theta_1)(\cos\theta_2 - \mathrm{i}\sin\theta_2)}{(\cos^2\theta_2 + \sin^2\theta_2)}$

$= \frac{r_1}{r_2}(\cos\theta_1 + \mathrm{i}\sin\theta_1)(\cos\theta_2 - \mathrm{i}\sin\theta_2)$

$= \frac{r_1}{r_2}(\cos\theta_1\cos\theta_2 - \mathrm{i}\cos\theta_1\sin\theta_2 + \mathrm{i}\sin\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2)$

$= \frac{r_1}{r_2}\big((\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2) + \mathrm{i}(\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2)\big)$

$= \frac{r_1}{r_2}(\cos(\theta_1 - \theta_2) + \mathrm{i}\sin(\theta_1 - \theta_2))$

So $\arg\left(\frac{z_1}{z_2}\right) = \theta_1 - \theta_2$

39)

$\arg\left(\frac{1}{z}\right) = \arg 1 - \arg z = 0 - \arg z = -\arg z$

Alternative Method:

Let $z = r(\cos\theta + \mathrm{i}\sin\theta)$

$\frac{1}{z} = \frac{1}{r} \times \frac{1}{\cos\theta + \mathrm{i}\sin\theta} = \frac{1}{r} \times \frac{1}{\cos\theta + \mathrm{i}\sin\theta} \times \frac{\cos\theta - \mathrm{i}\sin\theta}{\cos\theta - \mathrm{i}\sin\theta}$

$= \frac{1}{r} \times \frac{\cos\theta - \mathrm{i}\sin\theta}{\cos^2\theta + \sin^2\theta}$

$= \frac{1}{r} \times (\cos\theta - \mathrm{i}\sin\theta)$

Since $\cos\theta = \cos(-\theta)$ and $-\sin\theta = \sin(-\theta)$

$\frac{1}{z} = \frac{1}{r} \times (\cos(-\theta) - \mathrm{i}\sin(-\theta))$

So $\arg\left(\frac{1}{z}\right) = -\theta$

$\therefore \arg\left(\frac{1}{z}\right) = -\arg z$

40)

$\mathrm{i}z^2 = 9\mathrm{i}\left(\cos\frac{4\pi}{3} + \mathrm{i}\sin\frac{4\pi}{3}\right)$

$= 9\mathrm{i}\left(\cos\left(-\frac{2\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{3}\right)\right)$

Since $\mathrm{i} = \cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}$ then:

$\mathrm{i}z^2 = 9\left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)\left(\cos\left(-\frac{2\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{3}\right)\right)$

$= 9\left(\cos\left(\frac{\pi}{2} - \frac{2\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{3} + \frac{\pi}{2}\right)\right)$

$= 9\left(\cos\left(-\frac{\pi}{6}\right) + \mathrm{i}\sin\left(-\frac{\pi}{6}\right)\right)$

41)

$|z^6| = 729$ so $|z| = 3$

$\arg(z^6) = 6\arg z$ so $\arg z = \frac{5\pi}{18}$

$z = 3\left(\cos\frac{5\pi}{18} + \mathrm{i}\sin\frac{5\pi}{18}\right)$

So $a = 3\cos\frac{5\pi}{18}$ and $b = 3\sin\frac{5\pi}{18}$

42)

$z = 2\sqrt{3} + 2\mathrm{i} = 4\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)$

$w = -\sqrt{2} + \sqrt{2}\mathrm{i} = 2\left(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\right)$

$r: 4 \to 2$

θ: $\frac{3\pi}{4} - \frac{\pi}{6} = \frac{7\pi}{12}$

Enlargement by scale factor $\frac{1}{2}$

Rotation by $\frac{7\pi}{12}$ anticlockwise about the origin.

43)

$z = \sqrt{17}(\cos 1.3258\ldots + \mathrm{i}\sin 1.3258\ldots)$

$w = \sqrt{13}(\cos 2.55359\ldots + \mathrm{i}\sin 2.55359\ldots)$

$r: \sqrt{17} \to \sqrt{13}$

θ: $2.55359\ldots - 1.3258\ldots = 1.23$ (3 SF)

Enlargement scale factor $\frac{\sqrt{13}}{\sqrt{17}} = \frac{\sqrt{221}}{17} = 0.874$ (3 SF)

Rotate anti-clockwise 1.23 radians around the origin.

44)

$z = 2\sqrt{5}(\cos 0.4636 \ldots + \mathrm{i} \sin 0.4636 \ldots)$

$w = \sqrt{26}(\cos -1.3734 \ldots + \mathrm{i} \sin -1.3734 \ldots)$

$r : 2\sqrt{5} \to \sqrt{26}$

θ: $-1.3734 \ldots - 0.4636 \ldots = -1.84$ (3 SF)

Stretch scale factor $\frac{\sqrt{26}}{2\sqrt{5}} = \frac{\sqrt{130}}{10} = 1.14$ (3 SF)

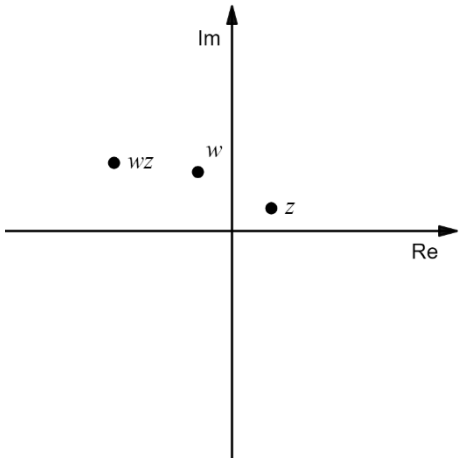
Rotate clockwise 1.84 radians around the origin.

45) a)

$z = 2\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)$

$w = 3\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right)$

$zw = 6\left(\cos\frac{5\pi}{6} + \mathrm{i}\sin\frac{5\pi}{6}\right)$

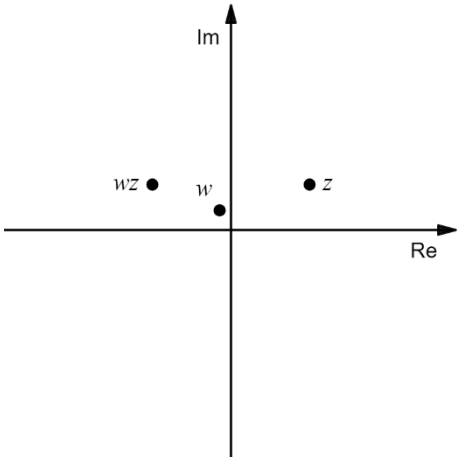


45) b)

$z = 4\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)$

$w = 1\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right)$

$zw = 4\left(\cos\frac{5\pi}{6} + \mathrm{i}\sin\frac{5\pi}{6}\right)$

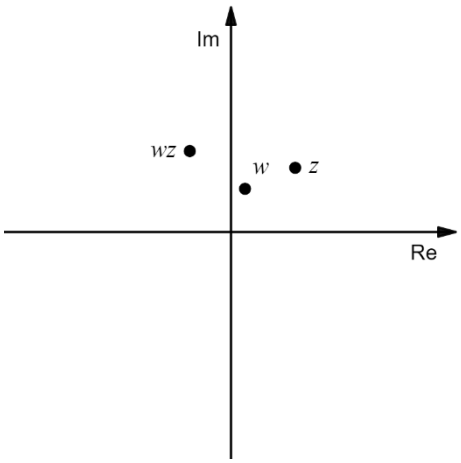


46) a)

$z = 2\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)$

$w = 1\left(\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}\right)$

$zw = 2\left(\cos\frac{13\pi}{20} + \mathrm{i}\sin\frac{13\pi}{20}\right)$

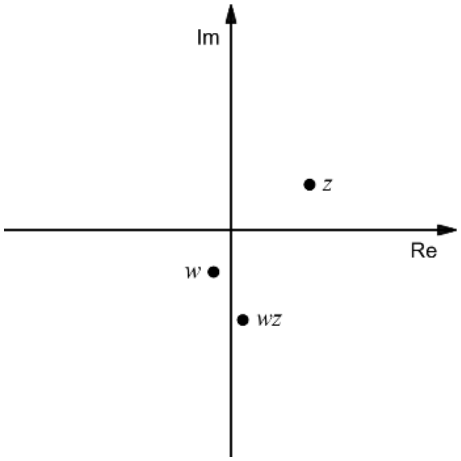


46) b)

$z = 2\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)$

$w = 1\left(\cos\left(-\frac{5\pi}{8}\right) + \mathrm{i}\sin\left(-\frac{5\pi}{8}\right)\right)$

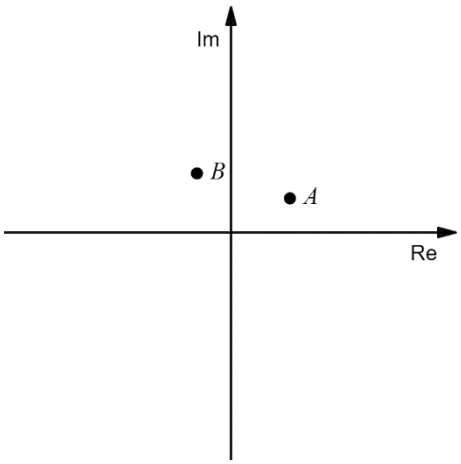
$zw = 2\left(\cos\left(-\frac{11\pi}{24}\right) + \mathrm{i}\sin\left(-\frac{11\pi}{24}\right)\right)$



47) a)

$z_1 = 3\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right) = \frac{3\sqrt{3}}{2} + \frac{3}{2}\mathrm{i}$

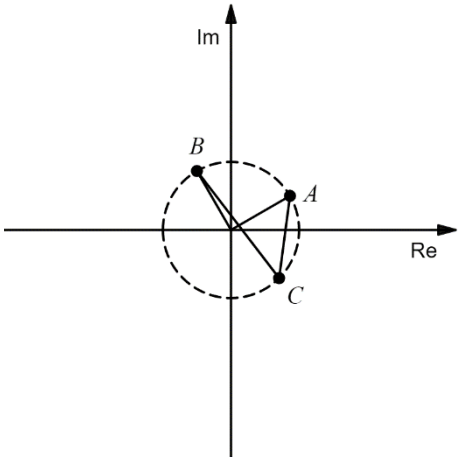
$z_2 = 3\left(\cos\left(\frac{2\pi}{3}\right) + \mathrm{i}\sin\left(\frac{2\pi}{3}\right)\right) = -\frac{3}{2} + \frac{3\sqrt{3}}{2}\mathrm{i}$



47) b)

All the points lie on a circle.

$\frac{2\pi}{3} - \frac{\pi}{6} = \frac{\pi}{2}$



Since the angle at the centre is twice the angle at the circumference, angle $\angle ACB = \frac{\pi}{4}$

48) a)

Let $z = a + b\mathrm{i}$

$|z|^2 = \left(\sqrt{a^2 + b^2}\right)^2$

$zz^* = (a + b\mathrm{i})(a - b\mathrm{i}) = a^2 + b^2$

$\therefore |z|^2 = zz^*$

48) b)

Let $z = a + b\mathrm{i}$ and $w = c + d\mathrm{i}$

$(z - w)^* = \left(a + b\mathrm{i} - (c + d\mathrm{i})\right)^*$

$= (a - c + (b - d)\mathrm{i})^*$

$= a - c - (b - d)\mathrm{i}$

$= a - b\mathrm{i} - c + d\mathrm{i}$

$= a - b\mathrm{i} - (c - d\mathrm{i})$

$= z^* - w^*$ as req.

48) c)

$|z - w|^2 = (z - w)(z - w)^* = (z - w)(z^* - w^*)$

$= zz^* - zw^* - wz^* + ww^*$

48) d)

$|z|^2 = zz^*$ so $zz^* = 2^2 = 4$

$|w|^2 = ww^*$ so $ww^* = 3^2 = 9$

$|z+w|^2 = (z+w)(z+w)^*$ so $4^2 = 16 = zz^* + wz^* + z^*w + ww^*$

$16 = 4 + wz^* + z^*w + 9$ so $wz^* + z^*w = 3$

$|z-w|^2 = zz^* - zw^* - wz^* + ww^*$

$= 4 - 3 + 9 = 10$

$\therefore |z-w| = \sqrt{10}$

49)

$|z+w|^2 + |z-w|^2$

$= (z+w)(z+w)^* + (z-w)(z-w)^*$

$= zz^* + zw^* + z^*w + ww^* + zz^* - zw^* - z^*w + ww^*$

$= 2zz^* + 2ww^* = 2|z|^2 + 2|w|^2$ as req.

Alternative Method:

Let $z = a + b\text{i}$ and $w = c + d\text{i}$

$|z| = \sqrt{a^2+b^2}$ and $|w| = \sqrt{c^2+d^2}$ $\therefore |z|^2 = a^2 + b^2$ and $|w|^2 = c^2 + d^2$

$|z+w| = \sqrt{(a+c)^2 + (b+d)^2}$ $\therefore |z+w|^2 = (a+c)^2 + (b+d)^2$

$|z-w| = \sqrt{(a-c)^2 + (b-d)^2}$ $\therefore |z-w|^2 = (a-c)^2 + (b-d)^2$

Now:

$|z+w|^2 + |z-w|^2 = (a+c)^2 + (b+d)^2 + (a-c)^2 + (b-d)^2$

$= a^2 + 2ac + c^2 + b^2 + 2bd + d^2 + a^2 - 2ac + c^2 + b^2 - 2bd + d^2$

$= 2a^2 + 2c^2 + 2b^2 + 2d^2$

$= 2(a^2 + b^2) + 2(c^2 + d^2)$

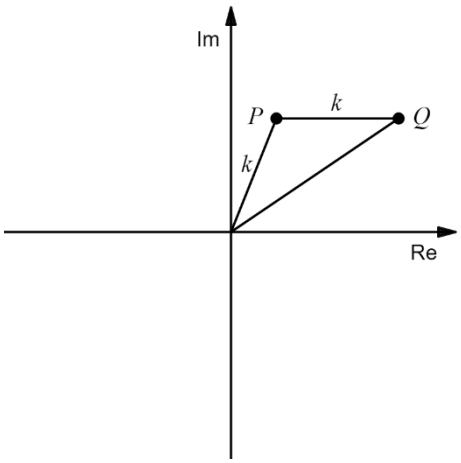
$= 2|z|^2 + 2|w|^2$ as req.

50) a)

$|z| = \sqrt{(k\cos\theta)^2 + (k\sin\theta)^2} = \sqrt{k^2(\cos^2\theta + \sin^2\theta)} = k$

50) b)

Sketch z and $z+k$ on an Argand diagram:



Let $\arg(z+k)=\phi$ so $\angle OQP=\phi$ and since this is an isosceles triangle, $\angle POQ=\phi$

$\therefore \arg(z+k)=\frac{1}{2}\arg(z)=\frac{1}{2}\theta$

51) a)

$\frac{z}{w}=\frac{2+2\sqrt{3}\mathrm{i}}{1+\mathrm{i}}\times\frac{1-\mathrm{i}}{1-\mathrm{i}}=\frac{2+2\sqrt{3}\mathrm{i}-2\mathrm{i}+2\sqrt{3}}{1+1}=\left(1+\sqrt{3}\right)+\left(-1+\sqrt{3}\right)\mathrm{i}$

51) b)

$|z|=\sqrt{2^2+\left(2\sqrt{3}\right)^2}=4$ and $\arg z=\arctan\left(\frac{2\sqrt{3}}{2}\right)=\frac{\pi}{3}$

$z=4\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)$

51) c)

$|w|=\sqrt{1^2+1^2}=\sqrt{2}$ and $\arg w=\arctan(1)=\frac{\pi}{4}$

$w=\sqrt{2}\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)$

51) d)

$\frac{z}{w}=\frac{4}{\sqrt{2}}\left(\cos\left(\frac{\pi}{3}-\frac{\pi}{4}\right)+\mathrm{i}\sin\left(\frac{\pi}{3}-\frac{\pi}{4}\right)\right)=2\sqrt{2}\left(\cos\frac{\pi}{12}+\mathrm{i}\sin\frac{\pi}{12}\right)$

51) e)

$(1+\sqrt{3})+(-1+\sqrt{3})\mathrm{i} = 2\sqrt{2}\left(\cos\frac{\pi}{12}+\mathrm{i}\sin\frac{\pi}{12}\right)$

Comparing imaginary parts:

$-1+\sqrt{3} = 2\sqrt{2}\sin\frac{\pi}{12}$

$\therefore \sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}}$ as req.

52) a)

$w = 4\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right) = 2\sqrt{2}+2\sqrt{2}\mathrm{i}$

52) b)

$\frac{z}{w} = \frac{4\sqrt{3}-4\mathrm{i}}{2\sqrt{2}+2\sqrt{2}\mathrm{i}} \times \frac{2\sqrt{2}-2\sqrt{2}\mathrm{i}}{2\sqrt{2}-2\sqrt{2}\mathrm{i}} = \frac{8\sqrt{6}-8\sqrt{2}\mathrm{i}-8\sqrt{2}-8\sqrt{6}\mathrm{i}}{8+8} = \left(\frac{\sqrt{6}}{2}-\frac{\sqrt{2}}{2}\right)-\left(\frac{\sqrt{6}}{2}+\frac{\sqrt{2}}{2}\right)\mathrm{i}$

52) c)

$|z| = \sqrt{\left(4\sqrt{3}\right)^2+4^2} = 8$ and $\arg z = \arctan\left(\frac{4}{4\sqrt{3}}\right) = -\frac{\pi}{6}$

$z = 8\left(\cos\left(-\frac{\pi}{6}\right)+\mathrm{i}\sin\left(-\frac{\pi}{6}\right)\right)$

52) d)

$8 \div 4 = 2$ and $\left(-\frac{\pi}{6}\right)-\frac{\pi}{4} = -\frac{5\pi}{12}$

$\frac{z}{w} = 2\left(\cos\left(-\frac{5\pi}{12}\right)+\mathrm{i}\sin\left(-\frac{5\pi}{12}\right)\right)$

52) e)

$\left(\frac{\sqrt{6}}{2}-\frac{\sqrt{2}}{2}\right)-\left(\frac{\sqrt{6}}{2}+\frac{\sqrt{2}}{2}\right)\mathrm{i} = 2\left(\cos\left(-\frac{5\pi}{12}\right)+\mathrm{i}\sin\left(-\frac{5\pi}{12}\right)\right)$

Comparing real parts:

$\frac{\sqrt{6}}{2}-\frac{\sqrt{2}}{2} = 2\cos\left(-\frac{5\pi}{12}\right)$

$\therefore \cos\left(-\frac{5\pi}{12}\right) = \frac{\sqrt{6}-\sqrt{2}}{4}$ as req.

53) a)

$\arg(wz) = \frac{\pi}{3} + \frac{\pi}{6} = \frac{\pi}{2}$

$\arg((wz)^2) = 2 \times \frac{\pi}{3} + 2 \times \frac{\pi}{6} = \pi$

$\arg((wz)^3) = 3 \times \frac{\pi}{3} + 3 \times \frac{\pi}{6} = \frac{3\pi}{2} \therefore \arg((wz)^3) = -\frac{\pi}{2}$

$\arg((wz)^4) = 4 \times \frac{\pi}{3} + 4 \times \frac{\pi}{6} = 2\pi \therefore \arg((wz)^4) = 0$

$\therefore n = 4$ and $|(wz)^4| = 2^4 \times 5^4 = 10000$

53) b)

$\arg(wz) = \frac{\pi}{6} + \frac{\pi}{2} = \frac{2\pi}{3}$

$\arg((wz)^2) = 2 \times \frac{\pi}{6} + 2 \times \frac{\pi}{2} = \frac{4\pi}{3} \therefore \arg((wz)^2) = -\frac{2\pi}{3}$

$\arg((wz)^3) = 3 \times \frac{\pi}{6} + 3 \times \frac{\pi}{2} = 2\pi \therefore \arg((wz)^3) = 0$

$\therefore n = 3$ and $|(wz)^3| = 3^3 \times 8^3 = 13824$

53) c)

$\arg(wz) = \frac{\pi}{4} - \frac{\pi}{12} = \frac{\pi}{6}$

$\arg((wz)^2) = 2 \times \frac{\pi}{4} - 2 \times \frac{\pi}{12} = \frac{\pi}{3}$

$\arg((wz)^3) = 3 \times \frac{\pi}{4} - 3 \times \frac{\pi}{12} = \frac{\pi}{2}$

$\arg((wz)^4) = 4 \times \frac{\pi}{4} - 4 \times \frac{\pi}{12} = \frac{2\pi}{3}$

Continuing this gives $\frac{5\pi}{6}, \pi, \frac{7\pi}{6}, \frac{4\pi}{3}, \frac{3\pi}{2}, \frac{5\pi}{3}, \frac{11\pi}{6}$

$\arg((wz)^{12}) = 12 \times \frac{\pi}{4} - 12 \times \frac{\pi}{12} = 2\pi \therefore \arg((wz)^{12}) = 0$

$\therefore n = 12$ and $|(wz)^{12}| = \sqrt{2}^{12} \times 3^{12} = 34\,012\,224$

54)

$\frac{1+\sqrt{3}\mathrm{i}}{2-2\mathrm{i}} \times \frac{2+2\mathrm{i}}{2+2\mathrm{i}} = \frac{(2-2\sqrt{3})+(2+2\sqrt{3})\mathrm{i}}{8} = \frac{1-\sqrt{3}}{4} + \frac{1+\sqrt{3}}{4}\mathrm{i}$

$\frac{1-\sqrt{3}}{4} + \frac{1+\sqrt{3}}{4}\mathrm{i} = \frac{\sqrt{2}}{2}\left(\cos\frac{7\pi}{12} + \mathrm{i}\sin\frac{7\pi}{12}\right)$

Consider $\left(\frac{\sqrt{2}}{2}\left(\cos\frac{7\pi}{12} + \mathrm{i}\sin\frac{7\pi}{12}\right)\right)^n$

Real means that the angle must be a multiple of π:

$\therefore n = 12$